

LABS-12: Value

DS1001 – Professor Wright

Due: 11:59pm day of lab (see Canvas)

Individual assignment

General Description: In this lab, you'll take on the role of a data scientist at a film studio to evaluate the performance of your movie score prediction model. You'll measure and interpret your model's performance using fairness metrics—True Positive Rate, False Positive Rate, Selection Rate, Demographic Parity Ratio, and Equalized Odds Ratio. By comparing results across the categories country, genre, and language, you'll evaluate whether the model works fairly for all types of movies and make a recommendation to the studio head about its use.

Why am I doing this? Data science does not end with building a model – it is your responsibility to make sure your model works fairly and reliably across different groups. In this lab, you'll practice identifying sources of bias and error in a model's predictions by examining how performance varies across movie types. You'll also learn how to use fairness metrics to evaluate whether a model is equitable, and how to explain your findings to someone without a technical background.

- **Course Learning Objective Alignment:** SWBAT identify common sources of bias and error in data and models across the ML pipeline.

What am I going to do?

1. Download the LABS-12_Value.ipynb file from Canvas and upload it to your GitHub Codespace that you prepared in LABS-03.
2. Activate the virtual environment you created (FirstnameLastname_env) **in the LABS-12_Value.ipynb file and the terminal** (if you forgot how to do this, check back at your notes and the instructions from that lab).
3. In the terminal, run the following commands. Press “enter” after each command.
a. `pip install fairlearn`
b. `pip freeze > requirements.txt`
4. Follow the instructions in the notebook to produce the needed output and answer the questions. Be sure to read through all of the markdown throughout the notebook and use the linked documentation when needed. *Remember: this is about understanding the process, not just running the cells. If you're not sure what something does, use your resources to find out – that is what labs are all about!*
5. **When you're done with the lab, make sure to save your work using the add/commit/push workflow that you learned in LABS-3.**

Questions

1. The metrics we will be using in this lab are **True Positive Rate**, **False Positive Rate**, **Selection Rate**, **Demographic Parity Ratio**, and **Equalized Odds Ratio**. For each metric:
 - a. Define the metric
 - b. Include any relevant equations
 - c. Define the range (i.e. what number represents complete fairness? Unfairness?),
 - d. Explain the situational implication of what this metric represents
2. For each category (**country**, **top_genre**, **top_lang**):
 - a. Are the classes in each category equally represented in the data?
 - b. If not, what class is favored?
3. Why is it important to consider class representation when evaluating the fairness of our model?
4. Report the overall metrics (**true positive rate**, **false positive rate**, **selection rate**) for the model.
5. For each category:
 - a. Report the **true positive rate**, **false positive rate**, and **selection rate** for the most prevalent & all other classes
 - b. Compare the metrics from part a to the overall metrics you reported in question 4. What does this comparison tell you about the model's performance for this class compared to others?
 - c. Report the **Demographic Parity Ratio** and **Equalized Odds Ratio**
 - d. Interpret the **Demographic Parity Ratio** and **Equalized Odds Ratio**. What does this suggest about the model's performance for this class compared to others?
6. Based on the fairness metrics you observed, do you think the model performs fairly for all movie types in the dataset? Why/why not?
7. Imagine you are a data scientist at a film studio. The studio head knows that higher-scoring movies often go on to win more awards. She asks if the model can be used to predict which movies will earn high scores and win awards **across all types of movies and award categories**, so she can guide advertising budgets. Based on your evaluation, make a recommendation to the studio head: should she rely on this model for that purpose or not? Support your recommendation with the metrics you used and be sure to explain any technical terms in plain language for a non-technical audience.

Tips:

- Remember the goal is to learn – not just to do. Expect to get stuck, poke around, look things up, and ask questions.
- Take advantage of your resources: your classmates, TA, the integrated copilot chat in VSCode, and the internet. Keep the first tip in mind – when you ask questions, do so to learn and understand, not just to do.
- Documentation can be intimidating at first. The more you read, the more comfortable it will become. Lean into the discomfort now – it pays off.

How will I know I have succeeded?

Specs Category	Specs Description
Formatting	• 3 page max • PDF format • Headings ◦ Lab Name ◦ Your name, course, date ◦ Questions ◦ Takeaway ◦ Notes
Questions	• Goal: Explore key elements of the data exploration process through the guided questions • Format your responses in a numerical list corresponding to the questions list • Make sure all screenshots are formatted as Figures (see below) and include all needed aspects described (i.e. your GitHub handle, code and output)
Takeaway	• Goal: Summarize the main things you learned or “takeaway” from this activity as it relates to the course • 1 paragraph (~ 5 sentences) • Use the formatting from the READ takeaways: a kernel that summarizes your takeaway in a few words, followed by a – and your explanation of the takeaway • Explicitly explain the connection between your takeaway from this activity, your personal growth, and the course/section.
Notes	• Goal: Create documentation for yourself to use when you repeat this process for your final project. • Any format that works for you

	• Include all supplemental information you think you may need. This could be instructions, links, or reminders about things you found confusing. • Remember you will be repeating this process independently in a few months. Create notes that future-you will be thankful for! • This section does NOT count toward your 3 page limit!
Figures	• Figures must be legible • Figures require a caption – e.g. Figure 1 – A chart showing GPU power consumption from 2000 to 2020. • Figures must be publication quality (i.e. this figure would not look out of place in a newspaper) • Use IEEE style (link) for captions. They should be ◦ Centered with the figure ◦ Below the figure

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